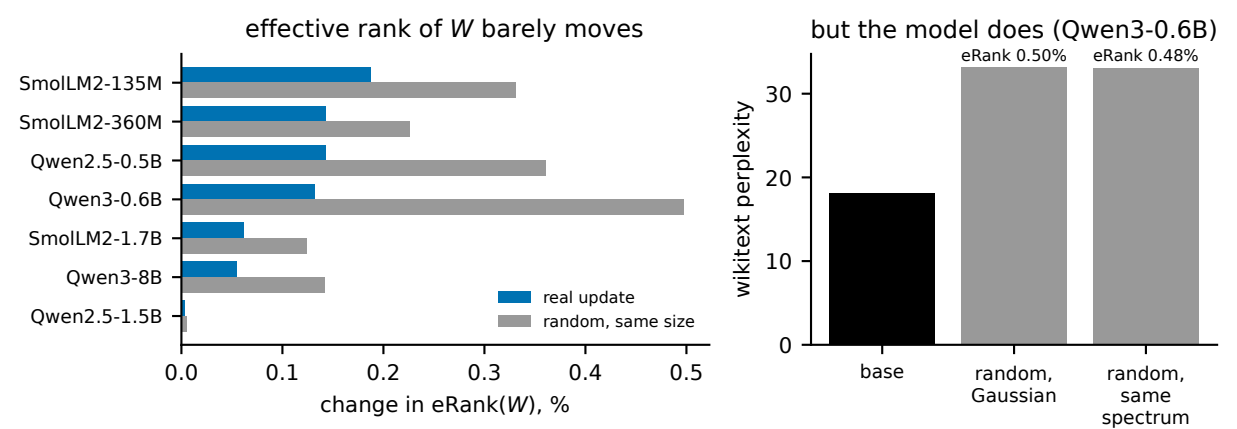
effective rank of W barely moves
SmolLM2-135M
SmolLM2-360M
Qwen2.5-0.5B
Qwen3-0.6B
SmolLM2-1.7B
Qwen3-8B
Qwen2.5-1.5B
real update
random, same size
0.0
0.1
0.2
0.3
0.4
0.5
change in eRank(W), %
but the model does (Qwen3-0.6B)
eRank 0.50%
eRank 0.48%
wikitext perplexity
0
10
20
30
base
random, Gaussian
random, same spectrum